Improper Integrals

Problem 1
Explain whether or not the following statement is true: if f is a continuous function and $\int_0^\infty f(x)\mathrm{d}x$ converges, then so does $\int_a^\infty f(x)\mathrm{d}x$ for all positive $a > 0$.

Solution.
This is true. Indeed, as the integral of a continuous function over a bounded set, $\int_0^a f(x)\mathrm{d}x$ is finite. This means that

$$\int_a^\infty f(x)\mathrm{d}x = \int_0^\infty f(x)\mathrm{d}x - \int_0^a f(x)\mathrm{d}x < \infty$$

as claimed. ■

Problem 2
Explain whether or not the following statement is true: if f is continuous and positive with $\lim_{x\to\infty} f(x) = 0$, then $\int_0^\infty f(x)\mathrm{d}x < \infty$.

Solution.
This is false. For instance $f(x) = \frac{1}{x}$ is positive and decays at infinity, but its associated improper integral diverges. ■

Problem 3
Compute

$$\int_1^\infty \frac{z}{(1+z^2)^3}\mathrm{d}z$$

Solution.
Fix a threshold $t > 1$. We use the change of variables $u = 1+z^2$,

$$\frac{\mathrm{d}u}{\mathrm{d}z} = 2z \Rightarrow \frac{1}{2}\mathrm{d}u = z\mathrm{d}z, \qquad z = 1 \to u = 2, \qquad z = t \to u = 1+t^2.$$

The integration by substitution formula gives

$$\int_1^t \frac{z}{(1+z^2)^3}\mathrm{d}z = \frac{1}{2}\int_2^{1+t^2} \frac{1}{u^3}\mathrm{d}u = -\frac{1}{4}\left[u^{-2}\right]_{u=2}^{u=1+t^2} = \frac{1}{16} - \frac{1}{4(1+t^2)^2}.$$

It follows that

$$\int_1^\infty \frac{z}{(1+z^2)^3}\mathrm{d}z = \lim_{t\to\infty}\int_1^t \frac{z}{(1+z^2)^3}\mathrm{d}z = \frac{1}{16} - \lim_{t\to\infty}\frac{1}{4(1+t^2)^2} = \frac{1}{16}.$$

■

Problem 4
Compute

$$\int_0^1 \frac{\ln(x)}{x}\mathrm{d}x$$

Solution.
Fix a threshold $0 < t < 1$. We can integrate by parts

$\pm$	D	I
$+$	$\ln(x)$	$\frac{1}{x}$
$-\int$	$\frac{1}{x}$	$\ln(x)$

to get

$$\int_t^1 \frac{\ln(x)}{x}\mathrm{d}x = \Big[(\ln(x))^2\Big]_{x=t}^{x=1} - \int_t^1 \frac{\ln(x)}{x}\mathrm{d}x = -(\ln(t))^2 - \int_t^1 \frac{\ln(x)}{x}\mathrm{d}x.$$

Rearranging yields

$$\int_t^1 \frac{\ln(x)}{x}\mathrm{d}x = -\frac{(\ln(t))^2}{2}.$$

It follows that

$$\int_0^1 \frac{\ln(x)}{x}\mathrm{d}x = -\frac{1}{2}\lim_{t\to 0}(\ln(t))^2 = -\infty,$$

so the integral diverges. ■

Problem 5
Compute

$$\int_3^\infty \frac{\mathrm{d}x}{x(\ln(x))^2}$$

Solution.
Fix a threshold $t > 3$. We use the change of variables $u = \ln(x)$,

$$\frac{\mathrm{d}u}{\mathrm{d}x} = \frac{1}{x} \Rightarrow \mathrm{d}u = \frac{1}{x}\mathrm{d}x, \qquad x = 3 \to u = \ln(3), \qquad x = t \to u = \ln(t).$$

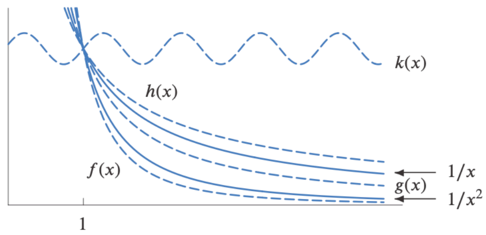
The integration by substitution formula gives

$$\int_3^t \frac{\mathrm{d}x}{x(\ln(x))^2} = \int_{\ln(3)}^{\ln(t)} \frac{1}{u^2}\mathrm{d}u = -\Big[u^{-1}\Big]_{u=\ln(3)}^{u=\ln(t)} = \frac{1}{\ln(3)} - \frac{1}{\ln(t)}.$$

It follows that

$$\int_3^\infty \frac{\mathrm{d}x}{x(\ln(x))^2} = \lim_{t\to\infty}\int_3^t \frac{\mathrm{d}x}{x(\ln(x))^2} = \frac{1}{\ln(3)} - \lim_{t\to\infty}\frac{1}{\ln(t)} = \frac{1}{\ln(3)}.$$ ■

Problem 6
Using the following graph decide whether the integral of each of the functions f, g, h and k on the interval $[1,\infty)$ converges, diverges or whether it is impossible to tell.



Solution.
Recall that the integral

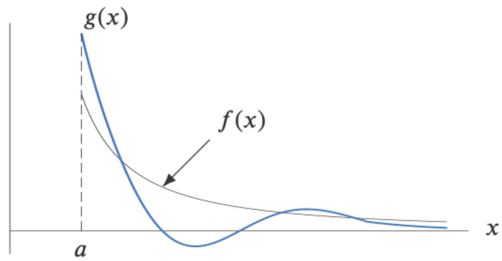
$$\int_1^{\infty} \frac{1}{x^p} \mathrm{d}x$$

converges if and only if $p > 1$. Since $f(x) \leq \frac{1}{x^2}$ for x large enough and $2 > 1$, the integral of f converges by the comparison test. The function $g(x)$ lies between $1/x$, whose integral diverges, and $1/x^2$ whose integral converges; it is therefore impossible to determine whether or not the integral of g converges. The integral of h diverges as h dominates $1/x$ whose integral diverges. The integral of k also diverges. Indeed, there exists a positive constant $c > 0$ with $k(x) > c$ for all $x > 1$. It follows by monotonicity of the integral that for any $t > 1$,

$$c(t-1) = \int_1^t c\mathrm{d}x \leq \int_1^t k(x).$$

Letting t tend to infinity establishes the claim. ■

Problem 7
Consider the functions f and g in the following graph.



If $\int_a^{\infty} f(x)\mathrm{d}x$ converges, what can you say about $\int_a^{\infty} g(x)\mathrm{d}x$?

Solution.
Recall that the convergence of an improper integral over an infinite interval depends only on the behaviour of the integrand for large values of x. For such values of x, the function f dominates the function g. Since the integral of f converges and g is continuous, it must be the case that the integral of g also converges. ■

Problem 8
Study the convergence of the improper integral

$$\int_1^\infty \frac{\mathrm{d}x}{1+x}$$

Solution.
Asymptotically, we have

$$\frac{1}{1+x} \sim \frac{1}{x}.$$

To see this, we can explicitly compute the limit of the ratio,

$$\lim_{x\to\infty} \frac{x}{1+x} = \lim_{x\to\infty} \frac{1}{1+\frac{1}{x}} = 1.$$

Since $\int_1^\infty \frac{1}{x}\mathrm{d}x$ diverges, the limit comparison test implies that $\int_1^\infty \frac{\mathrm{d}x}{1+x}$ also diverges. ■

Problem 9
Study the convergence of the improper integral

$$\int_1^\infty \frac{\mathrm{d}u}{u+u^2}\mathrm{d}u$$

Solution.
Asymptotically, we have

$$\frac{1}{u+u^2} \sim \frac{1}{u^2}.$$

To see this, we can explicitly compute the limit of the ratio,

$$\lim_{u\to\infty} \frac{u^2}{u+u^2} = \lim_{u\to\infty} \frac{1}{1+\frac{1}{u}} = 1.$$

Since $\int_1^\infty \frac{1}{u^2}\mathrm{d}u$ converges, the limit comparison test implies that $\int_1^\infty \frac{\mathrm{d}u}{u+u^2}$ also converges. ■

Problem 10
In Planck's Radiation Law, we encounter the integral

$$\int_1^\infty \frac{\mathrm{d}x}{x^5(e^{1/x}-1)}. \tag{1}$$

Argue that $1+t \leq e^t$ *for* $t \geq 0$, *and deduce that* (1) *converges.*

Solution.
Notice that $1+t$ is the tangent line to e^t at $t=0$. Since e^t is concave up, its tangent line is an under estimate so $1+t \leq e^t$ for all $t \geq 0$. Using this inequality with $t = 1/x$ shows that

$$1+\frac{1}{x} \leq e^{1/x} \Rightarrow \frac{1}{x} \leq e^{1/x}-1 \Rightarrow x^4 \leq x^5(e^{1/x}-1) \Rightarrow \frac{1}{x^4} \geq \frac{1}{x^5(e^{1/x}-1)}.$$

Since $4 > 1$ the comparison test gives the convergence of (1). ■